

EDUCATION

- **Iowa State University** Ames, IA
Ph.D. Student in Computer Science; GPA: 3.63/4.00 Aug. 2016 – 2021
- **University of Science and Technology of China** Hefei, China
B.S. in Modern Physics; Outstanding Graduated Student Aug. 2012 – July. 2016

PROGRAMMING SKILLS

- **Languages:** C/C++, Java, Python, Matlab, Verilog
- **Technology:** OpenCV, ROS, Gazebo, Linux Kernel

EXPERIENCE

- **Iowa State University** Ames, IA
Research and Teaching Assistant Aug. 2016 - 2021
 - **Research Assistant - Robotics Lab**
 - Physics analysis based dexterous robot manipulation and control: Currently working on picking up a kitchen knife with a multi-finger robot hand via finger gaiting and rolling (current research).
 - Developed a robotics communication middleware in modern C++ based on publisher-subscriber pattern and thread pool. Sensor and control messages are dispatched by a topic manager and processed by function objects.
 - Implemented a feature based image classifier using PCA and SVM.
- **Intel** Beijing, China
Linux Kernel Virtualization Intern Jun. 2018 - Sep. 2018
 - **KVM Linux Virtualization**
- **DJI** Shenzhen, China
Navigation Algorithm Intern Mar. 2016 - Jul. 2016
 - **Obstacle Sensing System**
 - A submodule of vision system in UAV, is used to indicate the obstacles on the flight direction and measure distance.
 - Obstacles are extracted on a stereo vision disparity map while tracking is performed by optical flow on obstacle feature points in main camera image. IMU and gimbal data map pixels between disparity image and main camera image.
 - The obstacle relative distance is estimated by a data fusion of UAV velocity and stereo vision measurement.
- **Conference reviewer experience**
 - International Conference on Intelligent Robots and Systems (IROS) Spring, 2018

SELECTED PROJECTS

RealTime Linux *kmap* Patch: A realtime Linux patch that enables kernel memory mapping function *kmap* to preserve mapping slots for realtime tasks and kill lowest priority task to release resource if necessary.

Mini CPU Design: Built a mini CPU based on single cycle PIC16F54 design by using a Xilinx FPGA. Assembly code and Verilog code co-design and co-verification, real testing in the FPGA board for simple applications (serial port).

Multi-Camera Visual SLAM: Developed a multi-camera SLAM ROS module based on RTABMap. 3D vision sensors can work collaboratively to build a map by merging frames in a centre node.

ShopEasy: An Android shopping list App with store navigation and recommendation system. Products can be automatically recommended based on the shopping list, user health information and recipe data from kaggle.com.

PUBLICATION

- Yan-Bin Jia and Yuechuan Xue. *Dexterous manipulation by two fingers with coupled joints*. IEEE International Conference on Robotics and Automation.
- Xiaoqian Mu, Yuechuan Xue, and Yan-Bin Jia. *Robotic cutting: mechanics and control of knife motion*. Submitted to IEEE International Conference on Intelligent Robots and Systems.

AWARD

- Robert Stewart Early Research Recognition Award, ISU Apr. 2018
- Iowa State University CS Hackathon, 3rd Place Feb. 2017
- Mathematical Contest in Modeling, Meritorious Winner (Top 10%~15%) Feb. 2016
- International Genetically Engineered Machine Competition, Software Track, Gold Award Oct. 2014